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ORIGINAL ARTICLE

Construction of a large language model for public demands integrating contextual knowledge reasoning and chain of thought

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Abstract

Efficient and accurate responses to public inquiries are critical for digital government services. However, existing rule-based and retrieval-based systems struggle with vague, multidimensional queries grounded in extensive policy documents, while large language models (LLMs) remain prone to knowledge gaps, topic drift, and hallucination. To address these limitations, we propose PD-LLM, a system that establishes real-time bidirectional coupling between LLMs and government knowledge graphs (KGs) via chain-of-thought (CoT) reasoning. At each reasoning step, PD-LLM dynamically retrieves and integrates policy facts, improving factual accuracy and coherence in multi-turn dialogues. Confidence-driven clarification and complexity-adaptive reasoning depth enable progressive transformation of vague user queries into structured entity queries, while quantitative stopping criteria ensure efficient termination upon information saturation. To enhance domain adaptability, the model employs two-stage fine-tuning on Qwen2.5-7B, strengthening its capacity to capture user intent, interpret policy content, and generate professional responses. Comprehensive automated evaluations demonstrate that PD-LLM consistently outperforms multiple baselines across n-gram precision, long-range dependency modeling, and semantic diversity. Dynamic KG injection further substantially reduces factual errors and authority overreach, yielding marked improvements in hallucination mitigation. These results confirm that integrating symbolic policy knowledge with stepwise LLM reasoning offers a practical and scalable solution for complex public service scenarios.

Keywords Large language model · Chain of thought · Public demands · Knowledge graph

Introduction

Addressing public demands has become a tangible reflection of the people-centered development philosophy within the context of new-era grassroots governance. These demands not only reflect critical social concerns but also serve as essential references for refining and adjusting the national governance system [1]. With the continuous advancement of e-government, the channels through which the public voices their demands have transitioned from traditional hotlines and petition systems to online platforms. As a result, online demands have become the primary form of public engagement in governance [2]. In an effort to provide more intelligent responses to the public's demands, the government is exploring the integration of LLM technology into online platforms [3]. Reports include the "Xuhui District Government Affairs Service Large Model" developed by the Shanghai Xuhui District Government and the "China Telecom Government Affairs Large Model" in Guangdong Province [4]. The introduction of LLMs is expected to significantly facilitate the digital transformation and efficiency optimization of grass-roots government affairs services. This will effectively improve the timeliness and accuracy of government affairs service responses and offer crucial theoretical underpinnings and practical approaches for constructing a diversified smart government affairs application system.

Traditional question-answering systems rely on rule-based methods, information retrieval, and shallow machine learning models. While these approaches offer strong interpretability within specific domains, they struggle with complex semantic structures and multi-turn dialogues [5]. Recent advances in generative AI, particularly LLMs such as ChatGPT, Claude, Deepseek, and LLaMA, have significantly

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improved question-answering performance across various sectors.Within the domain of government services, public demands constitute a critical component of social governance and governmental administration. On one hand, addressing these demands requires a deep understanding of policy texts. However, policy texts are inherently complex, characterized by their extensive length, structural intricacy, information density, content redundancy, and uneven distribution of implicit meanings. These features often pose substantial challenges for efficient information retrieval and accurate comprehension in policy-related question-answering scenarios, rendering it a highly complex information processing task [6]. Furthermore, the "inescapability" and "normativity" of policies during their implementation phase [7] necessitate that responses are not only accurate but also strictly adhere to established guidelines, adding another layer of complexity to response generation. On the other hand, most current online services still operate under an outdated "human-to-system adaptation" model, relying on fixed phrases or menu-driven operations. This forces the public to repeatedly adjust their often ambiguous or complex expressions before the system can clearly identify the target [8], highlighting the need for more intelligent and adaptive interaction mechanisms. The New Public Service Theory emphasizes that the governments should "serve rather than govern", placing greater focus on citizens' needs and public interests [9]. Government inquiry services must transition from passive information dissemination to actively addressing public demands. By leveraging LLMs, these services can better understand citizen intentions, contextualize demands comprehensively, and deliver authoritative, reliable solutions.

During inference, LLMs generate outputs based solely on parameterized knowledge. However, when encountering non-parameterized knowledge, they may produce inaccurate results that trigger erroneous reasoning chains, leading to hallucinations [10]. To address model hallucinations and generate high-quality responses, this study constructs a LLM for public demands that integrates downstream knowledge reasoning with a CoT mechanism. The model is termed the popular demands large language model (PD-LLM). By enabling dynamic interaction between the CoT and the KG, relevant governmental policy knowledge is incorporated at each stage of the reasoning process. Compared to traditional vector RAG and previous GraphRAG [11] approaches that follow a retrieve-and-generate framework of first building indexes then performing one-time summarization, PD-LLM's dynamic CoT-KG method deeply couples step-by-step reasoning for individual public requests with high-quality government affairs KGs. Each step of its reasoning updates knowledge based on the KG, and when information is insufficient, it drives multi-round questioning through KG topology. Moreover, the former primarily addresses global understanding problems over large-scale corpora and typically does not embed KG structures into step-by-step CoT reasoning and interactive Q&A processes. This research integrates KGs with fine-tuning to store policy texts at scale and enhance domain-specific performance. Tailored CoT templates are designed to improve the model's comprehension of public demands and grassroots service contexts. The contributions of this study are as follows:

1. We developed a government affairs knowledge enhancement framework that integrates CoT reasoning with KGs, enabling the model to more accurately comprehend and apply domain-specific knowledge in government affairs. The model demonstrates substantial improvements in understanding government affairs scenarios and acquiring professional knowledge. It can precisely capture user intentions and demands, while effectively enhancing the knowledge emergence capability of LLMs, thereby achieving comprehensive performance and efficiency optimization in government affairs knowledge Q&A services.
2. We designed innovative multi-level entity alignment mechanisms and domain-specific CoT templates tailored to the unique characteristics of the government affairs domain. These components enable precise semantic-level mapping and hierarchical association, ranging from basic administrative entities to complex policy concepts. This structured knowledge representation and reasoning framework significantly enhances the model's depth of semantic understanding and response accuracy in highly complex and nuanced government affairs scenarios.
3. By adopting a rigorous multi-faceted evaluation paradigm, the performance of PD-LLM was systematically compared with that of baseline models. This paradigm integrates traditional direct measurement indicators with the ability hierarchy framework in the field of scientific and technological intelligence, offering a comprehensive overview of the model's capabilities. Additionally, an in-depth case study was conducted to thoroughly validate PD-LLM's comprehensiveness and accuracy in addressing real-world, complex public demands.

Related studies

Fine-tuning of large language models

LLMs have demonstrated remarkable advantages in tackling complex tasks, owing to their exceptional natural language comprehension and text generation capabilities [12]. Even without explicit training for specific tasks, LLMs can

still manage various downstream tasks through appropriate prompting [13]. However, when dealing with vertical domain tasks that demand specialized knowledge, these models may suffer from hallucinations or provide inaccurate responses. To this end, fine-tuning techniques play a crucial role. By making refined adjustments on task-specific datasets, fine-tuning further enables the application of the broad knowledge acquired during the pre-training phase to concrete scenarios [14]. Consequently, numerous relevant LLMs have emerged in domains such as medicine [15], finance [16], and law [17]. Regarding government services, Emanuele et al. [18] fine-tuned LLMs and integrated them with multi-agent systems. This enabled LLMs to generate new documents in public management that adhered to the required structure. Although multi-agent systems can readily adapt to a vast amount of context, they increased the need for human supervision. Zhong et al. [19] fine-tuned a general model for the tax field based on the ChatGLM model using high-frequency real tax Q&A, but did not optimize the training data. Compared with ordinary LLMs, exhibits longer retrieval times.

The objective of fine-tuning is to retrain the model with labeled data, enabling the model to more effectively address problems within specific domains. Consequently, the quality of fine-tuning data exerts a decisive influence on the model's performance [20]. At present, however, most of the policy data suitable for LLM fine-tuning is collected and annotated manually. This approach is not only highly labor-intensive and time-consuming but also yields a limited quantity of data [21, 22]. As a result, the utilization of synthetic data as a supplement has emerged as a novel method for fine-tuning LLMs in vertical domains. Xiang et al. [23] proposed a retrieval-enhanced self-prompting method to construct a large-scale, high-quality question-answering dataset for science and technology policies. By generating synthetic data with detailed annotations, this approach reduces dependence on real data while mitigating privacy and copyright concerns. However, existing studies on domain-specific LLM fine-tuning predominantly rely on single data sources or lack effective knowledge integration mechanisms, limiting their capacity to address the diverse demands in government affairs [24]. Based on fine-tuning, this research adopts a two-stage fine-tuning strategy. The fine-tuning stage builds the model's deep understanding of professional terminology, logical frameworks, and administrative procedures through government Q&A, while instruction fine-tuning further refines its expression style to comply with official document writing conventions. This strategy more effectively integrates domain-specific expertise and policy response specifications, and simultaneously resolves the issue of a shortage of high-quality annotated data.

Knowledge reasoning and thought chain for enhancing large language models

KGs store a vast amount of fact in the form of triples, serving as a structured and deterministic means of knowledge representation. This explicit knowledge representation holds significant value in enhancing the knowledge perception capabilities of LLMs. In KG-enhanced LLMs, KGs can be incorporated not only into the pre-training and inference phases of LLMs to supply external knowledge but also utilized to analyze LLMs and offer interpretability [25]. For example, Graph RAG [26] constructs a graph-based text index from source documents and leverages this index to generate comprehensive and diverse responses to global queries, thereby enhancing the LLM's comprehension of the entire text corpus. LPKG [27] enables the LLMs to more effectively handle complex question-answering tasks involving retrieval by extracting data from KGs, generating reliable question-answering content. PoG [28] enhances the reasoning ability of the LLMs through KGs. By utilizing adaptive planning and self-correction mechanisms, the model can actively explore and store reasoning information, as well as rectify errors, thus improving the accuracy of knowledge retrieval, the reliability of reasoning, and the adaptability to complex problems. Although utilizing pre-built KGs and community summaries to answer global queries effectively addresses cross-textual and global perception issues, limitations remain in dynamic interaction and real-time knowledge updating [29]. Additionally, the lack of hierarchical and multi-turn interactive context means that while performance is good in terms of answer comprehensiveness and diversity, responses tend to be singular when facing real-time questions and users' multi-round follow-up inquiries [30]. To address this, PD-LLM integrates structured policy knowledge on-demand during its progressive reasoning process and automatically backtracks to original policy texts for supplementation when graph information is insufficient. Meanwhile, it leverages the entity-relationship topology of the KG to proactively initiate follow-up questions for progressive completion of critical information, thereby maintaining broad knowledge coverage while enhancing the system's real-time responsiveness to complex public demands, answer accuracy, and reasoning interpretability.

CoT is a prompting technique that facilitates LLMs in conducting complex reasoning by guiding them through intermediate reasoning steps. This technique has been shown to enhance the performance of LLMs in tasks encompassing mathematics, common sense, and symbolic reasoning [31]. The central concept of CoT involves decomposing a complex multi-step problem into multiple intermediate steps. This approach not only enhances the model's reasoning capabilities but also improves its interpretability. Nevertheless, CoT is susceptible to various issues, including calculation

errors, omission of reasoning steps, and semantic misinterpretations. In response to these challenges, Wang et al. [32] introduced PS Prompting. By implementing a "plan-then-solve" strategy, PS Prompting has significantly improved the reasoning quality, stability, and controllability of LLMs within zero-shot reasoning scenarios. However, due to the high sensitivity of LLMs to prompt formulation, this method necessitates iterative fine-tuning of prompts to identify the optimal prompt structure. Conversely, Kong et al. [33] proposed RP Prompting, which guides the style, tone, and focus of LLMs' responses by assigning specific roles to them. The interaction between LLMs and their assigned roles becomes more effective after alignment, enabling the satisfaction of specific requirements. Although RP Prompting triggers a more efficient form of implicit reasoning, its effectiveness is highly contingent upon the immersiveness Ness of role design and the selection of appropriate roles. To address the limitations of CoT when handling complex problems in knowledge-sparse domains, Vaidya et al. [34] proposed Chain of Concepts (CoC), which demonstrates significant performance in reducing hallucinations and enhancing the interpretability of reasoning. Inspired by Liu et al.'s [35] research on neural network hardware acceleration, this study draws upon the concepts of multi-level information fusion and progressive reasoning, applying them to the domain of knowledge inference in large language models. Distinct from quasi-neural network implementations at the physical layer, we have achieved multi-step decision optimization at the symbolic knowledge reasoning level through the integration of CoT mechanisms and dynamic KG integration.

Large language model regarding public demands

The LLM for Public demands is intended to enhance the response speed, precision, and personalization of government services by intelligently processing public appeals. It also aims to optimize decision-making support for government affairs and continuously adapt to and respond to the dynamic changes in public needs, offering 24/7 intelligent consultation services to the public [36]. In the process of public appeals expression, common challenges include ambiguous expressions, insufficient contextual understanding, limited capabilities in handling complex issues, as well as a lack of personalization and emotional understanding [37]. To address such issues, Hua et al. [38] developed a knowledge model based on the connotations of policy texts, which can effectively represent policy content and provide knowledge support for policy Q&A. This model effectively classifies policy-related questions by leveraging the optimized ERNIE + CNN, thereby offering accurate policy responses. Shen et al. [6] introduced LLMs into the field of policy information retrieval and proposed the Policy-RAG framework. This framework enhances the efficiency and effectiveness of policy text generation and retrieval by matching the semantic similarity between policy document knowledge bases and user questions. However, merely strengthening the retrieval and recognition of policy texts cannot guarantee the controllability of the generation results. Liu et al. [12] proposed an entity relation extraction model based on BERT-BiLSTM-CRF, which constructs a petition KG and vector index library, and combines knowledge reasoning and similar petition retrieval to guide the LLM to generate accurate responses to user petitions. However, this model assumes that users have "perfect" knowledge transmission capabilities and can clearly state their petitions from the very beginning, ignoring the problems of ambiguous expression and complex petitions. Han et al. [39] designed a domain-specific language model (GCALLM) for government consultation based on the daily consultation questions encountered by citizens. By injecting domain knowledge into the LLM and providing context information, the generated responses are more rigorous and accurate, but the model still exhibits hallucinations. Existing research on LLMs for public petitions still has deficiencies in handling complex petitions and policy interpretation, especially when users express ambiguously or have complex petitions, the model often has difficulty accurately understanding user intentions and providing precise responses. This study draws on the contextual reasoning approach of Han et al. and further integrates KGs and CoT to achieve more intelligent and personalized processing of government petitions, significantly improving the model's performance in complex scenarios. PD-LLM directly addresses these insufficiently explored issues through systematic adjustment and investigation, establishing a dynamic coupling mechanism that embeds knowledge retrieval within reasoning steps. Within PD-LLM's reasoning framework, the system maintains real-time interaction with the KG at every step of reasoning. When reasoning identifies specific policy constraints pertaining to a particular region, it can dynamically re-query supplementary policy provisions for that region, rather than being confined to the results of the initial retrieval.

Model framework

In this study, the LLM serves as the core component for event processing and dialogue reasoning. The overall framework of the designed PD-LLM is illustrated in Fig. 1. This research leverages the "12345" Government Service Platform in Jiangsu Province as its data source. Policy documents and public Q&A data are preprocessed using Data-Juicer. For policy texts, LLM-based prompt engineering automates entity-relationship extraction, followed by disambiguation, ontology alignment, and redundancy pruning. The refined triples are imported into Neo4j to construct a domain-specific

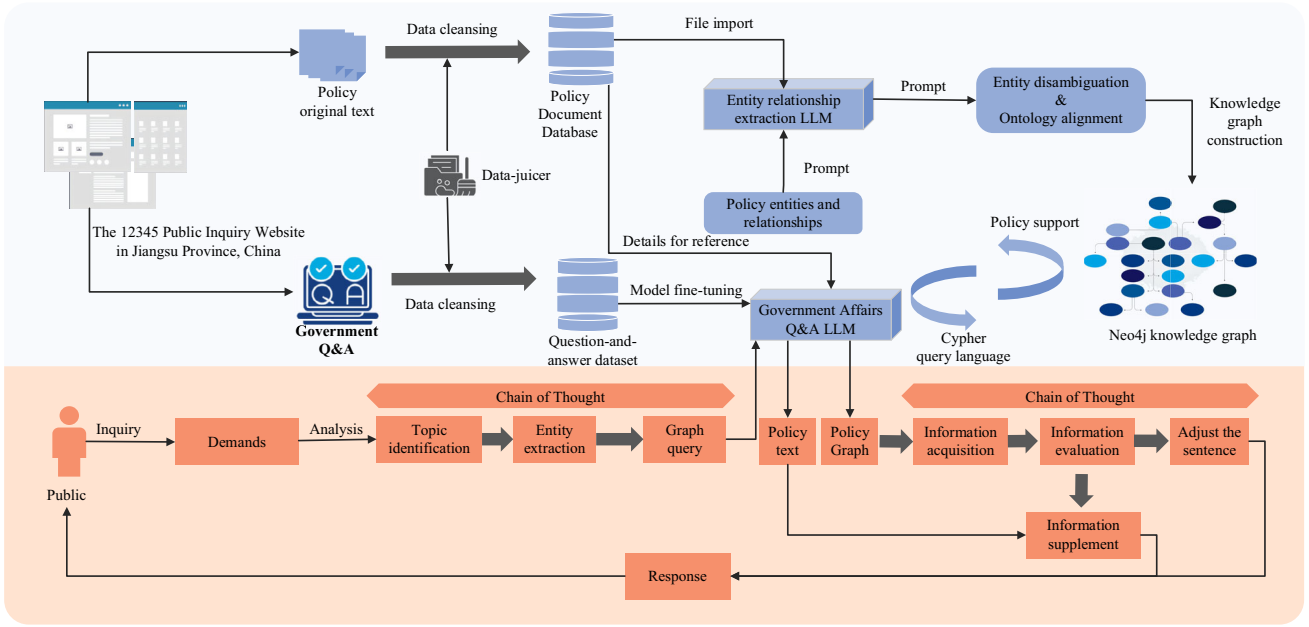


Fig. 1 The overall architecture of PD-LLM

Knowledge Graph. For Q&A data, the corpus is split into training and evaluation sets. The training set fine-tunes the LLM to embed government affairs response strategies, while the evaluation set assesses model performance. During inference, the model identifies domain entities in user queries, generates Cypher queries to interact with Neo4j, performs entity linking and knowledge reasoning, and generates contextually appropriate responses. Chain-of-Thought prompting is integrated throughout the inference pipeline to enhance reasoning quality.

The design framework of PD-LLM is established upon three interrelated theoretical assumptions that collectively constitute the theoretical foundation of the system's knowledge representation and reasoning mechanisms. (1) The KG representational capacity of explicit structured knowledge: The system assumes that explicit structured knowledge in the government affairs domain can be sufficiently represented through the node-edge structure of RDF triples. This assumption is grounded in the fundamental characteristics of KG representation. KGs adopt a "fact storage" paradigm and possess inherent advantages in representing first-order logic predicate relations [40, 41]. In the government affairs domain, this means that discretizable and explicit policy facts such as administrative procedures, conditional requirements, departmental contacts, and legal bases can be effectively encoded as graph structures. Furthermore, this provides the system with a structured, verifiable, and derivable foundation for knowledge storage and retrieval, enabling the LLM's reasoning process to no longer rely solely on parameterized knowledge but instead establish tight mapping relationships with external symbolic knowledge. (2) Progressive formalization of surface-level user needs: The system assumes that users' surface-level need expressions can be gradually transformed into formalized entity-relation queries through multi-turn clarification interactions. This assumption originates from the cognitive foundations of information retrieval and dialogue systems, where users' initial expressions often contain ambiguities or incompleteness, yet through systematic guiding questions, these needs can be progressively refined and structured [42]. Specifically, when classification confidence falls below the threshold, the system automatically generates clarification questions based on the entity-relation topology of the KG, leveraging the semantic association information inherent in the KG itself as the basis for clarification. (3) Alignment between CoT reasoning granularity and KG traversal: The system assumes that in typical single-domain government affairs problems, the semantic granularity required by CoT reasoning steps and the granularity of connected subgraphs returned by multi-hop KG traversal exhibit relative alignment at the statistical level. Knowledge in the government affairs domain possesses relatively regular hierarchical structures—single-entity problems correspond to one-hop direct relations, multi-entity problems require two-hop path expansion, and cross-departmental problems necessitate three-hop traversal. This enables the system to inject relevant symbolic knowledge through dynamic KG retrieval at each reasoning step, rather than passively depending on the LLM's parameterized knowledge.

Initially, the model receives public requests and performs a comprehensive analysis. By applying topic recognition techniques, it categorizes and identifies the type of inquiry.

Table 1 Government services knowledge graph entity type

ID	Entity type	Describe	Example
1	Service item	Specific government service programs	"Urban–rural resident pension benefits"
2	Government department	Government agencies providing services	"Human Resources and Social Security Bureau"
3	Service target	Types of people applying for services	"Individuals receiving the subsistence allowance"
4	Legal provision	Policy basis	Article 16 of the "Social Insurance Law"
5	Required material	Documents required for application	"ID card", "household register"
6	Process flow	Process of business handling	"Acceptance → Review → Issuance"
7	Service location	Location of service provision	"Municipal Government Service Center"
8	Administrative division	Geographical management scope	"Municipal-level", "County-level", "Town-level"
9	Time constraint	Time requirements for processing	"20 working days", "Immediate processing"
10	Acceptance condition	Qualification requirements for application	"Over 60 years old", "Local household registration"
11	Fee standard	Method of calculating cost amount	"Free", "50 yuan per time"
12	Approval result	Result after completion of processing	"Decision on Old-Age Insurance Benefits"
13	Service channel	Online/offline service mode	"Window service" and "online service"
14	Contact information	Information related to consultation or complaint	"12xxx", "0xx-xxxxxx902"
15	Other	Other precisely classified entities	"The entity subject to review of work performance"

Given a user query, the PD-LLM acts as a "government affairs domain expert classifier" to identify the primary welfare domain from a two-level hierarchical taxonomy. The first level comprises six macro-domains (Social Security, Housing & Real Estate, Transportation, Education, Health & Medical, Market Regulation), each further subdivided into fine-grained sub-categories aligned with the 12 entity types defined in the government affairs KG (Table 1). The model assigns a normalized confidence score to each candidate category via structured JSON-formatted output, ensuring machine-parseable results compatible with downstream KG operations. Following classification, a confidence-gated routing mechanism determines the subsequent inference path [43]. If the top-1 classification confidence exceeds a threshold $\tau = 0.80$, the query is routed directly to the corresponding domain-specific CoT template and KG subgraph, proceeding immediately to entity extraction. This high confidence threshold ensures that only semantically well-defined queries bypass clarification, reducing hallucination risks arising from ambiguous or under-specified inputs [44, 45]. Conversely, when confidence falls below τ, the system activates a multi-round clarification sub-routine. Leveraging the entity-relation topology of the KG, targeted supplementary questions are automatically generated (e.g., "Are you inquiring about urban residents' pension insurance or urban–rural residents' pension insurance?") to resolve semantic ambiguity and guide users to specify their intent incrementally, thereby progressively disambiguating complex requests before entering the main reasoning pipeline. The predicted category label directly conditions the selection of a domain-specific five-stage cascading prompt template. Each domain variant employs differentiated temperature parameters: $T = 0.1$ for initial Cypher query generation (favoring deterministic KG queries) and $T = 0.7$ for final answer generation (permitting greater expressiveness).

Subsequently, through entity extraction, the model detects key elements and government-related concepts. These extracted entities function as query nodes for multi-path retrieval within the KG, enabling the simultaneous and accurate retrieval of original policy texts. After obtaining the relevant information, the system not only evaluates the content to determine the completeness and authority of the response but also ranks the credibility of policy interpretations from various sources and resolves potential contradictions through this evaluation process. This ensures the consistency and reliability of the final responses generated.

Through the intelligent adjustment stage of response generation, the system dynamically modulates the level of professionalism and the mode of expression in its responses according to the user's profile and the scenario of their demands. Specifically, the intelligent information supplementation module, grounded in the earlier reasoning outcomes, proactively identifies latent information gaps. It then guides users to further specify their requirements or extracts supplementary explanations from relevant policies and regulations. This adaptive and cyclic reasoning model, which encompasses "identification- decomposition-excavation-refinement", not only enables the public to gradually clarify their implicit and often complex demands at each level but also prompts the model to continuously refine its comprehension of the associations between policies, regulations, and the issues at hand. Consequently, the model

generates solutions that are not only targeted and professional but also highly practical. This approach effectively elevates the intelligent level of government services and significantly enhances public satisfaction.

Construction of knowledge graph

In this research, the extraction of entities and relationships adheres to a three-stage pipeline consisting of "data pre-processing, LLM-guided extraction, and result post-processing". At the data pre-processing stage, specifically, long texts are segmented into units that are both semantically complete and compliant with the model's input constraints of the model. This is achieved through operations such as denoising, sentence segmentation, and semantic chunking, ensuring the preservation and traceability of contextual information.

Secondly, we design imperative prompts and restrictive schemas based on the LLM. Referring to approach by Ma et al. [46] built the policy KG, we clearly define 14 types of entities and 12 types of relationships. By employing structured prompt engineering and a few-shot learning approach, we guide the LLM to perform domain-specific entity recognition, covering 14 entity types such as service items, government departments, and legal clauses. Hierarchical extraction and filling techniques are applied to obtain multi-granularity structured information. To ensure data consistency, an entity deduplication mechanism based on hash caching and a unified entity index management system are introduced, enabling efficient handling of large-scale incremental data. During the KG construction phase, rule-based reasoning is utilized to automatically identify dependency relationships among entities based on 12 semantic relation types, resulting in a graph data format compatible with Neo4j. Furthermore, to ensure system stability and reliability, mechanisms such as state persistence, data validation, and exception handling are integrated into the process. Table 1 presents 14 types of entities along with their descriptions, and Table 2 shows 12 types of relationship types.

Given the widespread issues in complex multi-source heterogeneous government data, including the coexistence of numerous synonymous entities, inconsistent naming granularities, mixed use of abbreviations and full names, dynamic changes in department names, and inconsistent descriptive standards across different data sources [47], this study introduces an intelligent multi-level entity alignment mechanism following entity extraction to further enhance the standardization of entity representation, consistency of semantic associations, and reliability of subsequent knowledge reasoning in KGs. While this mechanism draws upon the ontology matching framework proposed by Euzenat and Shvaiko [48] in its overall approach, it does not adopt the single-layer lexical matching, rule constraints, or static similarity determination methods commonly found in traditional ontology matching techniques. Instead, considering the actual characteristics of government data being "multi-sourced, heterogeneously expressed, and semantically implicit," it constructs a four-tier progressive workflow of "dictionary alignment, similarity matching, deep semantic alignment, fallback processing," transforming entity alignment from single-pass determination into a process of hierarchical filtering, progressive confirmation, and dynamic supplementation.

At the first tier, the study constructs domain-specific standardized dictionaries covering core elements such as government departments, application materials, service targets, and policy items, prioritizing entities with clear semantic boundaries and relatively stable standardized names to achieve high-precision, low-ambiguity foundational alignment. This design demonstrates clear domain specificity, consistent with the importance of domain knowledge constraints and standardized expression emphasized by Ma et al. [35] in policy KG construction. Unlike traditional ontology matching that relies more heavily on general lexical resources or general rules, this step emphasizes the front-loaded embedding of government business knowledge into the alignment process, improving entity standardization quality from the source. The second tier employs edit distance and string similarity algorithms to identify entity variants, with a similarity threshold set at 0.8, handling cases such as abbreviations, alternative spellings, format differences, punctuation variations, and local word order changes, satisfying approximately 25% of alignment requirements. Compared to traditional methods that provide final results directly based on lexical similarity, this study treats this as an intermediate filtering layer, processing only surface-level differences within its applicable scope rather than extending to complex semantic scenarios, thereby reducing mismatching risks.

For entity relationships difficult to identify accurately through lexical similarity and rule constraints alone—such as implicit synonymy, hypernymy-hyponymy substitution, functionally equivalent expressions in policy contexts, or cross-departmental descriptive differences—this study introduces the advanced semantic understanding capabilities of large language models in the third tier, implementing deep semantic alignment combined with domain-specific prompt templates. The core of this tier is not merely determining whether two entities are "lexically similar," but analyzing their roles, functions, contextual constraints, and potential semantic correspondence relationships within specific governmental contexts, thereby improving recognition capabilities for complex heterogeneous expressions. This design receives indirect support from recent research: Trivedi et al. [49] demonstrated that combining retrieval mechanisms with CoT reasoning significantly improves processing effectiveness for knowledge-intensive tasks; Kang et al. [50] indicated

Table 2 Relationship types in the government service domain knowledge graph

ID	Relationship type	Direction of relationship	Relationship description	Example
1	Supervises	Department → Service item	The government department is responsible for managing a certain service	"Human Resources Bureau" supervises
2	Belongs_To	Department → Administrative region	The department belongs to a certain administrative level	"Human Resources Bureau" Belongs_To "Municipal"
3	Provides	Department → Service item	The department directly provides a certain service	"Government Affairs Center" provides "Document processing" services
4	Applies_To	Service item → Service recipient	The service is applicable to a specific group of people	"Minimum living allowance recipients" Apply_To "Low-income families"
5	Based_On	Service item → Legal clause	The legal basis for the service items	"Old-age insurance" Based_On "Social Insurance Law"
6	Requires_Condition	Service item → Acceptance conditions	The conditions that need to be met for handling the service	Requires_Condition "aged 60 or above"
7	Contains_Process	Service item → Processing procedure	The processing procedures of the service items	"Application" Contains_Process "Acceptance → Review"
8	Requires_Material	Service item → Processing materials	The materials required for handling the service	Requires_Material "Household register"
9	Charges_Fee	Service item → Cost situation	The cost standard of the service	Charges_Fee "200 yuan"
10	Handled_At	Service item → Processing location	The service can be handled at a specific location	"Handled_At" "Window processing"
11	Time_Limit	Service item → Time limit	The time limit for handling the service	Time_Limit "Five working days"
12	Contact_Via	Department → Contact information	The consultation methods with the department	"Human Resources and Social Security Bureau" Contact_Via "12xxx"

that incorporating KG information into language models helps enhance models' knowledge awareness and semantic understanding capabilities. Compared to existing ontology matching techniques, the key improvement of this study lies precisely here: traditional methods typically excel at handling explicit lexical, structural, or logical correspondence relationships, while this study further strengthens adaptability to complex contexts, implicit semantics, and ambiguous expressions.

Following the first three tiers, the system establishes a fallback processing mechanism for supplementary mapping and result consolidation of entities that cannot achieve high-confidence alignment, avoiding impacts on the overall structural coherence of the KG due to unaligned long-tail entities. This also reflects another important difference from traditional ontology matching techniques: traditional methods often focus more on local judgments of "whether matching occurs," while this study simultaneously concerns itself with "whether the KG remains structurally complete, referentially consistent, and sustainably maintainable after matching." Therefore, during entity standardization, the system completely preserves mapping relationships and alias information between original and standardized entities, automatically updating all entity references in relationship files to ensure alignment occurs not only at the entity level but is synchronously reflected at the relationship and graph structure levels. Thus, entity compression, semantic unification, and structural consistency maintenance are integrated into the same workflow, both enhancing the traceability of alignment results and providing support for subsequent knowledge reasoning, graph updates, and error auditing.

In terms of actual effectiveness, this multi-level entity alignment strategy compressed the initial 103,132 entities to 87,409 standardized entities, achieving a compression rate of 15.25%, demonstrating significant success in reducing entity redundancy, improving standardization levels, and enhancing semantic consistency. Overall, the proposed method differs from existing ontology matching techniques not merely by adding a large language model component, but by forming a comprehensive alignment framework oriented toward government scenarios that encompasses hierarchical processing,

Table 3 Government service entity alignment pseudocode

Algorithm 1 Entity Alignment

Input: Original entity file, entity type mapping

Output: The entity file after alignment

1. **Initialization:** Aligner (LLM), alignment cache { }, ID mapping table { }, standard entity library { }, standard dictionary ()
2. **for** each (file name, entity type) in the entity file mapping **do**
3. Original entity list ← Load CSV file, Alignment result list ← []
4. **for** each entity in the original entity list **do**
5. Entity name ← Entity['name'], Cache key ← Entity type + ":" + Entity name
6. Align entities ← {

Cache hit: Align cache if cache key in align cache
Dictionary match: Dictionary alignment (entity name, entity type) if entity name in standard dictionary [entity type]
Similarity match: Similarity alignment (entity name, entity type) if similarity match (entity name, standard entity library) > threshold
AI alignment: Parse AI result (call large model (construct alignment prompt (entity name, entity type))) OR create fallback entity (entity name)}

7. Update cache and mapping: alignment cache[cache key] ← alignment entity, ID mapping table[entity['id']] ← alignment entity['id'], alignment result list + = alignment entity
8. **end for**
9. Process and save: Remove duplicate entity list ← Remove duplicates based on standard names, save file
10. **end for**
11. **Generate Output:** Save the file, update the relationship file. The result report ← Generate a processing statistical report

dynamic semantic understanding, result fallback processing, and end-to-end traceability. Compared to traditional ontology matching approaches that emphasize general rules and static similarity calculations, this mechanism demonstrates stronger adaptability, accuracy, and engineering controllability in complex multi-source heterogeneous government data environments. The pseudocode for this process is shown in Table 3.

If there are still entities remaining after the above process, manual proofreading will be adopted for entity disambiguation. Through continuous efforts, a comprehensive and structurally complete KG in the field of government affairs has been gradually constructed.

Response enhancement dataset

In order to steer LLMs to generate responses that are more in line with the norms of policy Q&A and user requirements, this research constructed a policy response enhancement dataset consisting of 360 data items. This dataset was based on the basis of high-quality policy Q&A data. It was constructed by manually annotating JSON files, with GPT-4 utilized as an auxiliary means to guarantee the accuracy and consistency of the annotations. The annotation format of this synthetic dataset is < Question, Department, Region, Category, Policy, Answer > . Here, Question represents the actual government consultation question raised by the user; Department is the specific government department responsible for the business; Region is the geographical area involved in the question; Category is the category of public welfare it belongs to; Policy is stored in the form of an array, containing the source and specific content of the policy; Answer is the reply content. Eventually, 360 data samples were constructed. An example of this dataset is shown in Table 4.

Table 4 Examples of enhanced response datasets

Field name	Field example
Question	How to apply for a replacement of a lost vehicle registration certificate?
Department	Haimen District Public Security Bureau of Nantong City
Region	Nantong City
Category	Traffic and Transportation\Vehicle Management\Driver's License Management
Policy	[Source: Replacement of Lost Vehicle Registration Certificate] If you have a vehicle photo, bring it along. If you don't have a vehicle photo, the vehicle owner should drive the original vehicle to the vehicle management office to take a photo for the registration certificate and then go to the window of the vehicle management office with your ID card to apply for…
Answer	Hello! Regarding the issue of replacing a lost vehicle registration certificate, we will now provide you with a detailed explanation of the application process: I. Preparation of Required Materials The materials needed vary depending on whether you have kept a photo of your vehicle that meets the requirements: 1. If you have saved a vehicle photo that meets…

Although the volume of high-quality manually annotated data is limited (360 samples), it plays a vital role in guiding models to learn intelligent and standardized response strategies for government affairs. This dataset effectively mitigates generalization errors and irrelevant responses that LLMs may generate during policy interpretation, particularly when handling complex and nuanced policy content. From a transfer learning perspective, a small set of carefully curated high-quality data can lead to substantial performance gains in specific tasks, especially in domains such as government services, where accuracy and standardization are critical for intelligent decision-making. Certainly, expanding the dataset size represents a valuable direction for future research. While maintaining current quality standards, the data scale can be increased through semi-automated annotation methods or by

incorporating policy Q&A samples from additional regions and categories, thereby further enhancing the model's intelligent adaptability across diverse and evolving government service scenarios.

Fine-tuning of the model

In this study, Qwen2.5-7B, an open-source LLM from Alibaba Cloud, is chosen as the base model for PD-LLM. Recognizing that directly applying a general LLM to the task of addressing complex public demands may result in inaccurate responses or "hallucination" phenomena due to a lack of domain-specific knowledge, the LoRA (Low-Rank Adaptation) technique is employed. This advanced parameter-efficient fine-tuning approach is crucial for transforming the general LLM into an intelligent domain-specific expert. Its core principle involves keeping the original weight parameters of the pre-trained model in a frozen state while embedding additional low-rank matrix decomposition layers within the model architecture. Only these newly introduced trainable parameters are optimized and updated [38], allowing for efficient adaptation to the specialized and often complex knowledge requirements of government affairs [51].

Figure 2 shows that for the pre-trained weight matrix $W_0 \in R^{d\times d}$, LoRA constrains its update mechanism. Specifically, the incremental parameter matrix ΔW in full-parameter fine-tuning is expressed as the low-rank approximation of two matrices with fewer parameters, namely A and B:

$$W_0 + \Delta W = W_0 + BA \tag{1}$$

Here, $B \in R^{d\times r}$ and $A \in R^{d\times r}$ are the weight matrices for LoRA low-rank adaptation, with rank r being much than the original dimension d. During training, the parameters W_0 are frozen, and only the parameters in A and B are trained. As shown, for $h = W_0x$, the forward propagation process becomes:

$$h = W_0x + \Delta Wx = W_0x + BAx \tag{2}$$

Therefore, LoRA can not only significantly reduce the computational cost and storage overhead of the fine-tuning process, but also achieve comparable performance to full-parameter fine-tuning on specific tasks.

A two-stage fine-tuning strategy—domain fine-tuning followed by instruction fine-tuning—is adopted to customize the LLM. Domain fine-tuning leverages 29,933 structured government affairs Q&A pairs to build foundational capabilities, enabling the model to understand professional terminology, policy frameworks, and service logics. Instruction fine-tuning then refines the model's expressive capabilities using 360 instruction examples with rich metadata, calibrating response style, structure, and tone to align with professional government service standards.

Dynamic integration of thought chains and knowledge graph

In order to tackle the problem of non-parametric knowledge emerging during question-answering, this research utilizes CoT to guide every step of the problem-solving procedure. It dynamically retrieves and supplements the knowledge necessary for the corresponding steps. For a single conversation, given the demand D, the model first identifies the entities existing in D according to the intention of the question. Meanwhile, it prompts the LLM to quantitatively rank these entities in terms of relevance based on the question intention. Special attention is paid to the topic entities with high relevance, and they are mapped and linked to the corresponding nodes in the KG. Subsequently, the topic extraction step is executed.

The LLM is prompted to evaluate E and the emerging entities via a structured scoring template. For each candidate entity $e_i \in \mathcal{E}^{cand}$, the LLM is explicitly instructed to assign a relevance score $s(e_i, E) \in [0,1]$ along two dimensions: (1) semantic relevance to the core intent of E, and (2) type compatibility, i.e., whether the entity type matches one of the expected types defined in Table 1 given the query category. The prompt template is formalized as:

> "For each extracted entity, rate its relevance to the query on a scale of 0 to 1, considering: (a) how central it is to the query's core intent [0–0.6], and (b) whether its entity type matches the expected type for this query category [0 or 0.4]. Output a JSON list of {entity, score} pairs."

The composite score is thus:

$$s(e_i, D) = s_{sem}(e_i, E) + s_{type}(e_i, E),\ s_{sem} \in [0,0.6],\ s_{type} \in \{0,0.4\} \tag{3}$$

The topic entity set is then determined by a fixed threshold $\tau = 0.5$:

$$\mathcal{E}_0^{topic} = \{e_i \in \mathcal{E}^{cand} \mid s(e_i, E) \geq \tau\} \tag{4}$$

The cardinality $N = \mid \mathcal{E}_0^{topic} \mid$ is thereby determined by the thresholded output rather than by an unconstrained LLM decision, while the LLM's contextual understanding is preserved as the primary scoring mechanism.

To impose a principled upper bound on the reasoning depth and to allocate computational resources in proportion to query complexity, we introduce a query complexity-adaptive reasoning depth allocation strategy. Specifically,

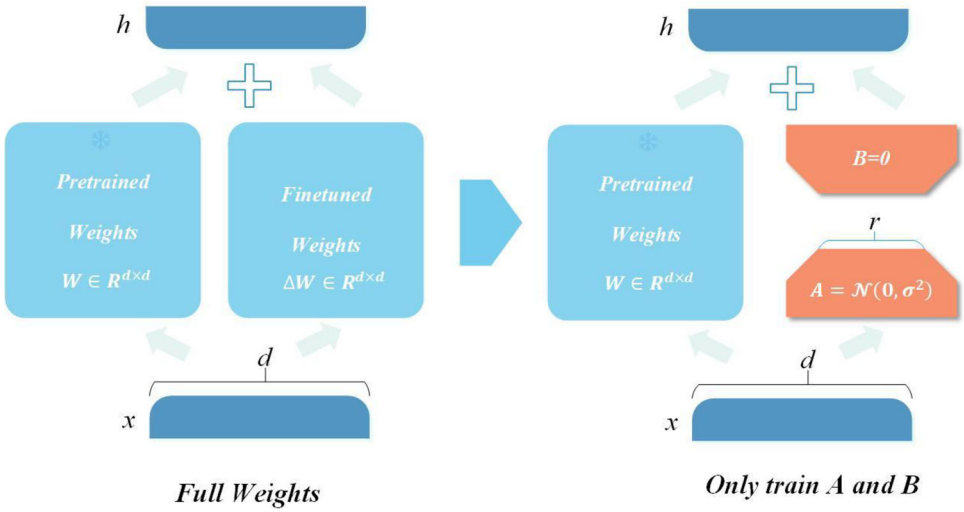


Fig. 2 LoRA weight processing

two structural features are computed at the entity recognition stage: the topic entity count N, and a cross-department indicator 1_{cross}, which is set to 1 if the identified Government Department-type entities are affiliated with distinct administrative hierarchies as defined by the Belongs_To relation in the KG1, and 0 otherwise. Based on these two features, the maximum reasoning step T_{max} is assigned as follows:

$$T_{max} = \begin{cases} 1, \text{ if } N = 1 \wedge 1_{cross} = 0 \\ 2, \text{ if } 2 \leq N \leq 3 \wedge 1_{cross} = 0 \\ 3, \text{ if } N \geq 4 \vee 1_{cross} = 1 \end{cases} \tag{5}$$

This assignment reflects the intrinsic topological structure of the constructed KG: single-entity queries can be resolved via direct one-hop traversal along relations such as Contact_Via or Time_Limit; multi-entity queries require two-hop path expansion to bridge semantically related but structurally separated nodes; cross-department queries necessitate three-hop traversal to aggregate distributed policy information across administrative boundaries. Thereafter, graph retrieval is conducted. Descriptive information is extracted from policy entities that are topically relevant to the initial topic entities, and other adjacent nodes are accessed.

At each reasoning step t, the system applies a composite stopping criterion to determine whether to proceed with further graph traversal. Let $\mathcal{E}_{new}(t)$ denote the set of entities newly retrieved at step t, and $\mathcal{C}_{t-1}$ denote the accumulated context entity pool up to step $t-1$. The stopping condition is formally defined as:

$$\text{Stop}(t) = 1[|\mathcal{E}_{\text{new}}(t) \setminus \mathcal{C}_{t-1}| < \delta] \vee 1[t \geq T_{max}] \tag{6}$$

The first term constitutes an information saturation condition: if the number of newly retrieved entities that are not already present in the accumulated context falls below a threshold δ (empirically set to 2 in this study), continued traversal is deemed to yield diminishing informational returns, and the reasoning loop terminates early. This condition replaces the previously implicit reliance on the LLM's subjective self-assessment of information sufficiency with an objective, graph-native quantitative criterion that incurs no additional model inference overhead. The second term constitutes a hard upper-bound condition: regardless of whether the information saturation condition is triggered, the reasoning loop terminates unconditionally upon reaching the pre-assigned step limit T_{max}, ensuring that the computational complexity of the entire inference pipeline remains strictly bounded. When $Stop(t) = 1$, if the accumulated context $\mathcal{C}_t$ remains insufficient to construct a complete response, the LLM additionally reads policy documents associated with $\mathcal{E}_0^{topic}$ and extracts salient passages as supplementary evidence.

At each step of the reasoning process, the model fuses the structured knowledge retrieved from the KG with the current reasoning state in real time, ensuring that every decision-making stage is based on the most relevant and up-to-date policy information. Meanwhile, when the initial information is still insufficient to construct a complete solution, the model intelligently designs subsequent interaction questions based on the entity-relation topology of the KG to guide users to provide supplementary information, achieving multi-round progressive problem-solving. Through this dynamic integration mechanism of the CoT and KG, PD-LLM can maintain the logical rigor of the reasoning process while continuously injecting domain-specific expertise, significantly enhancing the model's accuracy, authority, and interpretability in parsing complex government affairs issues.

In Fig. 3, in response to the user's query "What is the housing purchase subsidy standard for high-level innovative talents in the urban area of Nantong City?", the LLM first

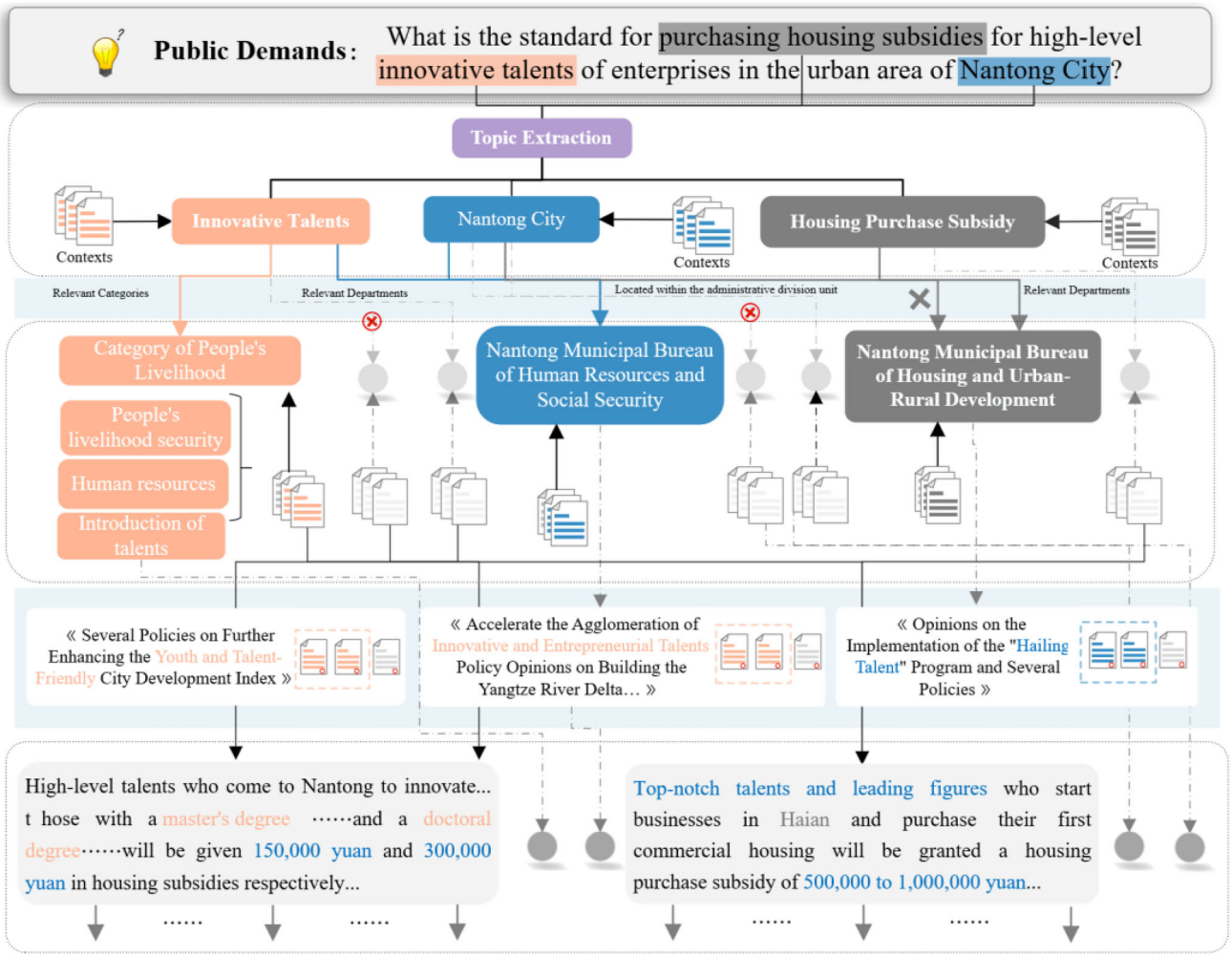


Fig. 3 An example of the dynamic integration of chain of thought and knowledge graphs

conducts entity recognition and extraction through prompt engineering, successfully identifying key candidate entities, including the geographical entity "Nantong City", the talent category entity "innovative talents", and the policy entity "housing purchase subsidy". Subsequently, the model queries the KG, systematically collecting node information related to each candidate entity and their semantic adjacency relationships, forming a structured context knowledge pool. This example corresponds to a Type-2 query under the query complexity-adaptive reasoning depth allocation strategy ($N = 3$, $1_{cross} = 0$, $T_{max} = 2$): the first reasoning step traverses the Applies_To relation to identify the applicable talent categories, and the second step traverses the Charges_Fee relation to retrieve the specific subsidy standards, yielding a complete two-hop inference path of the form $e_0 \xrightarrow{r_1} e_1 \xrightarrow{r_2} e_{answer}$. These associated nodes include policy category classification ("social welfare category"), relevant government functional departments ("Nantong Human Resources and Social Security Bureau", "Nantong Housing and Urban–Rural Development Bureau"), and other multi-dimensional information. Based on this context knowledge pool rich in semantic relationships, the LLM can perform refined KG query reconstruction, accurately locating complex policy information, and thus efficiently retrieving specific policy provisions and standards related to housing purchase subsidies for high-level innovative talents. When the LLM detects semantic ambiguity or missing key information in the request, making it impossible to accurately extract entities or locate relevant policies, it will trigger a multi-round dialogue interaction strategy. Based on the entity relationships defined in the KG, targeted questions are designed to guide the user to gradually supplement the missing information elements or resolve semantic ambiguity points. This process will then be repeated.

Prompt engineering

Prior to the five-stage cascading QA pipeline, a dedicated classification prompt is applied to determine the inquiry category. The classification prompt is structured as follows: the LLM is assigned the role of an expert in government affairs classification; it is provided with the predefined taxonomy as a structured reference schema; it is instructed to output a JSON object with fields primary_domain, sub_category, confidence, and ambiguity_flag; and it is given three in-context exemplars (few-shot examples) covering representative query types across distinct welfare domains. This few-shot classification strategy is consistent with the prompt engineering approach described by Wei et al. [31], where emergent classification capabilities of LLMs are elicited through structured in-context learning rather than dedicated fine-tuning of a separate classifier. The classification result directly parameterizes the Cypher query template used in Stage 1 of the five-stage pipeline, ensuring that graph traversal is initiated from the most semantically relevant KG subgraph for the identified domain.

In the entity extraction module, to ensure comprehensive information extraction from policy documents by the LLM, prompt engineering is designed as a structured information extraction framework. Through the development of a JSON template and a context-aware mechanism, this approach enables the precise transformation of unstructured text into structured entities within the KG. Initially, the LLM is assigned a specific role, referred to as a "professional assistant for structured extraction of government service information." Subsequently, the prompt strategy is adapted based on whether the input text originates from a long document. If the input is identified as a text segment, additional instructions are included, guiding the model to extract recognizable information within that segment while leaving incomplete or missing fields blank. Specifically, for service item entities, core attributes such as name, type, and description are extracted. For auxiliary entities, associated information is retrieved across multiple dimensions, including government departments, service recipients, legal bases, and processing procedures. Finally, a well-defined JSON format is established to standardize the LLM's output. This approach not only improves the accuracy and efficiency of information extraction but also aligns with the requirements of academic research in data processing and knowledge representation.

In the domain of government affairs question-answering, this study proposes a multi-stage prompt engineering framework that establishes a five-stage cascading processing paradigm. The framework comprises five stages: (1) translating natural language queries into Cypher statements to access the Knowledge Graph; (2) dynamically refining query paths based on preliminary results; (3) converting structured outputs into natural language responses; (4) integrating multiple knowledge sources through a fusion mechanism to enrich retrieved information; and (5) extracting precise knowledge from policy documents. Differentiated temperature parameters (0.1, 0.2, and 0.7 for respective stages) balance precision and fluency, while a rigorous inter-stage information transfer mechanism ensures seamless integration from query understanding to answer generation. By combining the structural advantages of Knowledge Graphs with policy document content, this framework delivers accurate, comprehensive, and user-centric government affairs consultation services.

Experiment

Experimental settings

Dataset

The policy data used in this study is sourced from Jiangsu 12345 (https://12345.jszwfw.gov.cn/cns-bmfw-wsbsdt/index/pages/default/index2.html), a platform hosted by the Jiangsu Provincial Data Bureau. This platform includes policy Q&A and government documents from 13 prefecture-level cities in Jiangsu Province. The data covers 19 areas of public welfare, involving departments such as the Science and Technology Bureau, Public Security Bureau, Civil Affairs Bureau, Justice Bureau, Labor and Social Security Bureau, Environmental Protection Bureau, Transportation Bureau, Agriculture and Forestry Bureau, Tourism Bureau, and Health Bureau at both the provincial and prefecture-level city levels in Jiangsu.

The data are mainly divided into two categories: policy text data and policy Q&A data. Policy text data include regulations, policy documents, administrative orders and other formal policy texts issued by the government. These texts usually go through rigorous legal and policy-making procedures and have high objectivity and authority. Policy Q&A data, on the other hand, cover the questions and answers between the public and policy makers regarding policy-related issues, presented in a question-and-answer format, with more diverse content and a certain degree of subjectivity. A total of 185,552 pieces of data were collected. Figure 4 shows the types and regional distribution of Q&A data.

The 12345 government service hotline Q&A data in Jiangsu Province exhibits significant regional imbalance. By volume, Suzhou (20,426), Lianyungang (10,601), and Yancheng (6237) dominate, reflecting differentiated regional demands. Suzhou, as an economic hub, leads in economic integration (11,115), agriculture/forestry/water conservancy

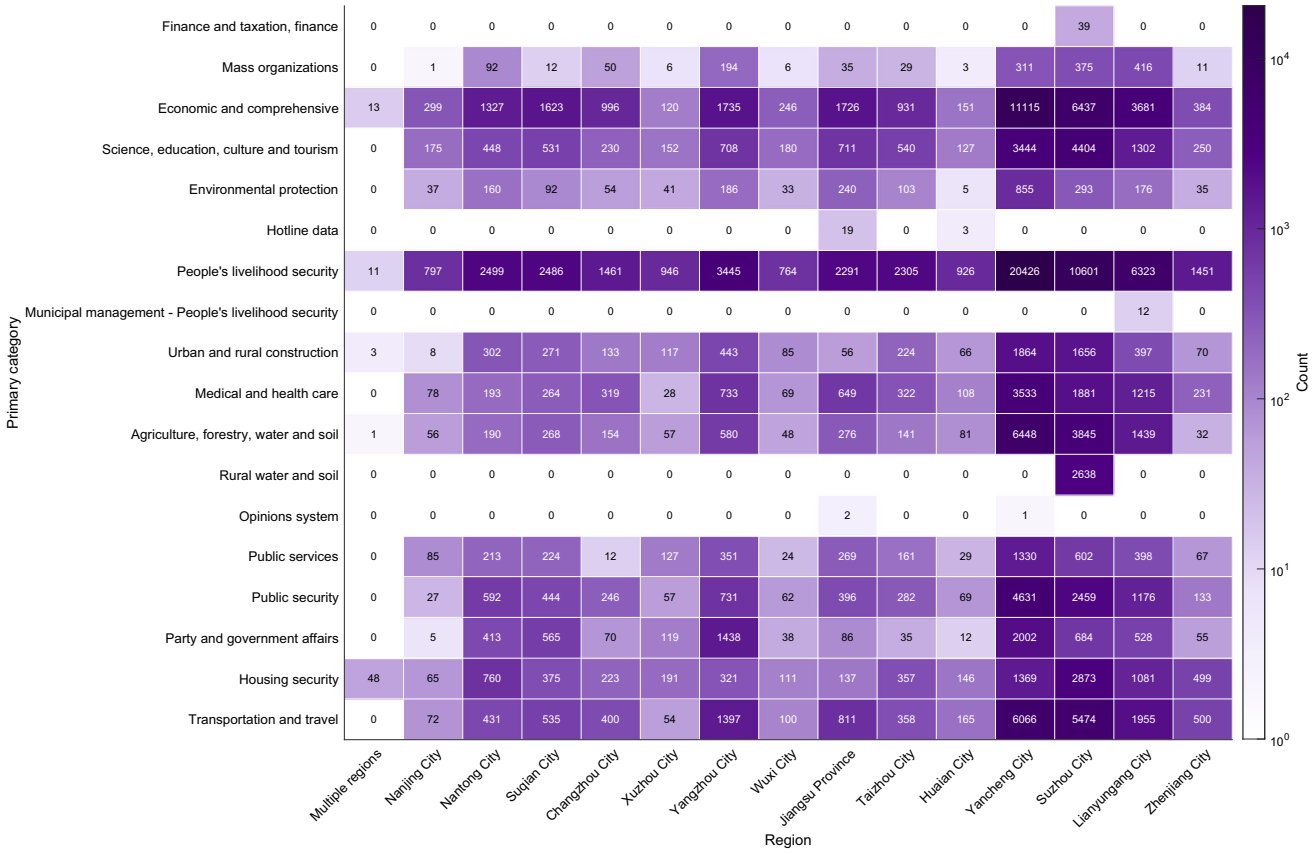


Fig. 4 Distribution of government affairs Q&A data

(6448), and transportation (6066)—aligning with its development profile. Lianyungang shows distinct structural features, with prominent concerns in agriculture/forestry/water conservancy (2638) and science/education/culture/tourism (4404), reflecting its industrial characteristics. Provincially, economic integration, transportation, and public security issues predominate, indicating livelihood priorities. Conversely, justice and finance/taxation queries are sparse or absent across most cities, revealing disparities in public participation and institutional accessibility.

Based on this, Data-Juicer, a one-stop multi-modal data processing system, was used to remove meaningless characters, filter out low-quality samples, identify and delete samples with excessive repetition, eliminate abbreviated language expressions, unify text formats, and conduct manual verification of the cleaned data to ensure its accuracy and completeness. Ultimately, 33,259 policy Q&A and 15,607 policy text data were obtained. In the research, the policy Q&A were divided into training and evaluation sets in an 8:2 ratio, and the policy text was used to construct a KG.

Evaluation metrics

BLEU (Bilingual Evaluation Understudy) is centered on the n-gram precision. Here, n-gram refers to the number of consecutive words. The basic idea is to count the occurrence of each n-gram in the machine translation result within the reference translation and take the ratio of this count to the total number of n-grams in the machine translation as the precision. The formula is as follows: BP is the brevity penalty factor; p_n is the n-gram precision; w_n is the weight coefficient, usually set as $w_n = \frac{1}{N}$; N is the maximum length of n-gram. According to n-gram, it can be divided into various evaluation indicators, and the common ones are BLEU-1, BLEU-2, BLEU-3, and BLEU-4.

$$\text{BLEU} = \text{BP} \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{7}$$

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) is an indicator used to evaluate automatic text summarization and machine translation. ROUGE-n is a similarity evaluation metric based on n-grams (Eq. 4), and ROUGE-L is a similarity evaluation metric based on the

longest common subsequence (Eq. 5). Meanwhile, in the ROUGE evaluation metrics, -f, -p, and -r represent different calculation methods or focuses. -r emphasizes the coverage of the generated text to the reference text. -p focuses on the accuracy and relevance of the generated text. -f is a balance of the two, with weights adjusted according to specific needs.

$$\text{ROUGE} - \text{N} = \frac{\sum_{S \in Ref\,Summaries} \sum_{\text{gram}_n \in S} \text{Count}_{\text{match}}(\text{gram}_n)}{\sum_{S \in Ref\,Summaries} \sum_{\text{gram}_n \in S} \text{Count}(\text{gram}_n)} \quad (8)$$

$$\text{ROUGE} - \text{L} = \frac{(1+\beta^2) \cdot R_{\text{lcs}} \cdot P_{\text{lcs}}}{\beta^2 \cdot P_{\text{lcs}} + R_{\text{lcs}}} \quad (9)$$

METEOR (Metric for Evaluation of Translation with Explicit Ordering) is a metric employed to evaluate the quality of machine translation outputs. METEOR takes into consideration a broader range of factors, including synonym matching, stem matching, and word order.

$$\text{METEOR} = F_{\text{mean}} \times (1 - \text{Penalty}) \quad (10)$$

$$F_{\text{mean}} = \frac{10 \times P \times R}{R + 9 \times P},\ P = \frac{m}{w_t},\ R = \frac{m}{w_r},\ \text{Penalty} = \gamma \times \left(\frac{ch}{m}\right)^{\theta} \quad (11)$$

The three traditional metrics, BLEU, ROUGE, and METEOR, provide fundamental means for quantifying text similarity when evaluating the generation performance of LLMs, measuring the degree of conformity between the generated text and the reference text from aspects such as precision, recall, and semantic matching.

Hallucination detection

Current hallucination detection metrics predominantly adopt single-dimensional designs (such as n-gram overlap, embedding similarity, and NLI methods), which struggle to adequately capture the complexity of hallucinations. A large-scale benchmark study by Kulkarni et al. [52] demonstrates that GPT-4 exhibits superior performance in contextual understanding and multi-factor reasoning, with its judgments showing significantly higher consistency with human evaluation compared to traditional methods. Furthermore, government administrative responses exhibit highly standardized and structured characteristics, involving multi-dimensional benchmark comparisons—including verification of policy basis accuracy, precise delineation of authority boundaries, and correct application of conditional stipulations—complex relationships that single similarity metrics cannot adequately capture. Consequently, drawing on their approach, we employ GPT-4 as an evaluation agent, leveraging its powerful contextual reasoning capabilities to construct a structured assessment system based on a five-option framework.

Within this framework, Option A (subset—information with lower volume but flawless accuracy), Option C (expression equivalence—identical content with different wording), and Option D (direct contradiction—apparent factual error) possess relatively clear judgment standards, representing binary determinations amenable to logical verification. Option B (exceeding authority or misrepresentation—altering original policy intent) involves the gray zone of degree judgment, requiring contextual consideration of whether changes constitute substantive alteration, thus warranting particular attention. Option E (expression differences without substantive relevance) is most critical, profoundly reflecting the unique reality of government Q&A scenarios. Authoritative responses in government Q&A are typically provided directly by government departments, conventionally employing verbatim policy restatement, strictly adhering to official document language norms, presenting highly standardized, templated, and formulaic expression styles that prioritize terminological accuracy and semantic rigor, though relatively lacking in practicality and readability. Conversely, content generated by large language models undergoes refinement, optimization, and restructuring in practical application, and while preserving the core policy meaning, typically employs more fluent and accessible phrasing that renders complex policy clauses more comprehensible and implementable for ordinary citizens. Such expression-level differences do not equate to factual errors or policy deviation. In actual government service scenarios, the primary objective of citizens accessing policy information is understanding rights and obligations, clarifying administrative procedures, and grasping application conditions—substantive content rather than verbatim replication of government terminology. Therefore, Option E's design purpose is to accurately distinguish two core concepts: optimization of expression form versus substantive alteration of policy meaning. When differences involve only wording choices, statement sequence, information organization methods, or redundancy adjustment—-expression dimensions—they fall under Option E; whereas when differences involve expansion or contraction of authority scope, addition or deletion of applicable conditions, or changes to core meaning—substantive dimensions—they should be classified as Option B or D. The proportion of model-generated responses free from harmful hallucinations (HFR) serves as the final evaluation metric:

$$\text{HFR} = \frac{N_A + N_C + N_C}{N_{total}} \quad (12)$$

N_A is the number of class A samples, N_C is the number of class C samples, N_C is the number of class E samples, and N_{total} l is the total number of samples.

Table 5 summarizes and clarifies all the evaluation indicators.

Table 5 Evaluation metrics summary

Metric	Meaning	Explanation
BLEU-1 ~ 4	n-gram precision	Measure exact matching of 1-g, 2-g, 3-g, and 4-g respectively. Higher orders capture longer semantic units
ROUGE-1-p/r/f	Unigram-level evaluation	p = precision (accuracy of generated content), r = recall (coverage of reference content), f = harmonic mean of both
ROUGE-2-p/r/f	Bigram-Level Evaluation	Same as above but based on consecutive 2-word combinations
ROUGE-L-p/r/f	Long-range coherence	Based on Longest Common Subsequence (LCS), evaluating overall text structural coherence and sequence preservation
METEOR	Semantic equivalence	Considers synonyms, stems, and word order simultaneously, transcending lexical-level matching
GPT4-HFR	Harmful hallucination ratio	Uses GPT-4's five-category framework (A: accurate, B: exceeding authority, C: expression equivalence, D: direct contradiction, E: expression differences) to assess content accuracy

Implementation details

A large-scale parallel computing architecture was established by utilizing six NVIDIA GeForce RTX 4090 graphics cards. Each card is equipped with 24GB of GDDR6X video memory, resulting in a total of 144GB of GPU video memory. This setup was integrated with an Intel Xeon Platinum 8352V processor featuring 32 virtual cores and a base frequency of 2.10GHz. The system was further configured with 240GB of high-speed memory, leveraging the PyTorch 2.1.0 deep-learning framework. It was based on the Ubuntu 22.04 LTS operating system and incorporated the CUDA 12.1 unified computing architecture. All experiments involving language models were carried out under identical hyperparameter settings for training, validation, and testing. The fine-tuning process employed LoRA with a rank of 258 and alpha of 512, targeting all linear layers. The hyperparameters were configured as follows: the learning rate was set at 2×10^{-5}, the total number of training epochs was 10, the gradient accumulation steps were set to 4, and the warm-up ratio was 0.05. For each conversation, the maximum number of output tokens was set to 2048. The values of top_p and temperature were set at 0.8 and 0.6 respectively. The overall training process consumed approximately 20 h. The hyperparameter settings specified above (learning rate 2×10^{-5}, training epochs 10, etc.) applied exclusively to the fine-tuned models in the two-stage fine-tuning process. In contrast, the four baseline models (Llama3.1-8B, Qwen2.5-7B, DeepSeek-R1-7B, and Mistral-7B) were evaluated in their original pre-trained state without any fine-tuning intervention.

Experimental results and analysis

The experimental results adopted a multi-dimensional evaluation paradigm, integrating traditional direct measurement indicators with the ability hierarchy framework in the field of science and technology intelligence, to form a multi-dimensional and multi-level evaluation method for LLMs. This method, while retaining the quantitative characteristics of formalized measurement tools, introduced ability-oriented evaluation dimensions, constructing a complete evaluation spectrum from micro-level language expression to macro-level ability performance.

Qwen2.5-DFT is a domain fine-tuning version, using 29,933 government-related QA pairs for domain knowledge adaptation. Qwen2.5-IFT is an instruction fine-tuning version, using 360 augmented data instances to optimize response format and compliance. FT-LLM is the model fine-tuned through a two-stage process. FT + CoT integrates a CoT reasoning mechanism on top of two-stage fine-tuning, whereas FT + KG only integrates the KG retrieval capability. PD-LLM is the model that, on the basis of fine-tuning, incorporates KGs and CoT, and the other baselines are all open-source LLMs. Specifically, Llama is the latest open-source LLM released by Meta; Qwen is a LLM launched by Alibaba Cloud; DeepSeek is a high-performance LLM released by DeepSeek; Mistral is an open-source LLM launched by the French artificial intelligence company Mistral AI. Vector-RAG projects policy documents into vector space and creates indexes. Upon receiving queries, it retrieves relevant documents using vector similarity metrics, then augments the query with these retrieved results as external knowledge before inputting them into a large language model. GraphRAG, developed by Microsoft, leverages hierarchical Leiden community detection to partition KGs into multi-level communities and generates summaries for each. Using a Map-Reduce framework, it concurrently generates intermediate answers from community summaries, aggregates them through priority sorting, and produces a final answer—enabling complex reasoning and handling of global questions. LightRAG [53] converts entities and relationships into Key-Value indexes stored in a vector database. It employs a dual-layer retrieval strategy: low-level retrieval

performs precise entity and edge lookups based on concrete keywords, while high-level retrieval searches for relationships and their endpoint nodes using abstract keywords. All three approaches utilize models fine-tuned through a two-stage process for inference, with consistent parameter configurations across other settings.

To verify the statistical significance of performance differences among models across various evaluation metrics, this study employs independent samples t-tests [54] to conduct significance testing on the comparison results ($\alpha = 0.05$). In Tables 6 and 7, a standard significance marking scheme is used to annotate the test results1: a single asterisk (*) indicates $p < 0.05$, reaching the significance level; double asterisks (**) indicate $p < 0.01$, reaching the highly significant level. Unmarked values indicate that the difference from the comparison baseline does not reach statistical significance.

Analysis of indicator evaluation

The results of the metric evaluation are shown in Table 6 and Table 7. From the macroscopic performance of the fifteen automated metrics, PD-LLM demonstrates a significant comprehensive advantage with the best scores in thirteen metrics, especially leading in key dimensions such as BLEU-1 to BLEU-4, high-order ROUGE-L, and Meteor, which measure lexical coverage, long-range dependency, and semantic diversity. In contrast, the four 7B/8B-level open-source baseline models trained solely on instruction data lag behind in all core metrics, with the most obvious disadvantages in high-order n-gram consistency and document-level matching, indicating that relying solely on native capabilities is insufficient to meet the refined language requirements of complex tasks.

Supported by 29,933 government affairs domain question–answer pairs, Qwen2.5-DFT demonstrates significant performance improvements. ROUGE-1-p increased from 0.1519 in the base model to 0.3513, while ROUGE-1-r rose from 0.4076 to 0.4571, and ROUGE-2-r climbed from 0.1251 to 0.2242. This improvement indicates that through targeted domain-specific fine-tuning, the model is able to learn the unique expression patterns and conceptual relationships within the government affairs domain. The consistent growth across BLEU metrics further confirms that n-gram matching has improved at multiple levels. These findings support an important observation: in professional domain applications, high-quality domain data plays a foundational role in enhancing model performance, and this improvement is not merely a matter of parameter adjustment, but rather an elevation of the model's capacity for deep understanding of domain knowledge.

The instruction fine-tuning stage presents an interesting phenomenon. Although FT-IFT uses only 360 augmented data samples, which appears extremely limited compared to the data scale of the previous stage, it maintains a high ROUGE-1-r of 0.4589 while showing a slight decline in ROUGE-1-p. This seemingly contradictory observation reveals the essential difference between instruction fine-tuning and task-oriented fine-tuning. The goal of instruction fine-tuning is not to achieve absolute gains on existing metrics, but to improve the model's understanding of task intent and its adherence to normative expression conventions within the government affairs context. This finding carries important practical implications: in professional domains, a small amount of high-quality instruction data often promotes model adaptability and normativity more effectively than large-scale undifferentiated data.

Without the intervention of external knowledge, pure supervised fine-tuning of the baseline model can bring considerable gains: FT-LLM outperforms the un-fine-tuned model by 3 to 8 percentage points in the entire BLEU series and ROUGE-1's f and p directions. Among them, the ROUGE-1-p score rises to 0.4637, leading among the models in the same table. This phenomenon indicates that fine-tuning significantly enhances the model's ability in local literal matching and short-range information alignment, and enables it to more accurately reproduce the key expressions in the reference answers.

FT + KG produced improvements across multiple dimensions. The gains in ROUGE-L-r and ROUGE-L-f indicate that external knowledge structures enable the model to better maintain long-range semantic coherence, which is particularly important for government affairs question answering that requires multi-step reasoning. The improvement in BLEU-4 reflects that relational information in the KG helps the model generate more complex and well-formed expressions. Notably, this improvement is accompanied by a slight decline in precision, which is not a flaw in the system but rather an inherent characteristic of knowledge augmentation—the introduction of external knowledge inevitably expands the model's response scope, resulting in a certain trade-off in local literal matching precision. This finding suggests that when evaluating knowledge-augmented systems, one should not focus excessively on a single precision metric, but should instead consider the overall balance between coverage and consistency.

The standalone application of FT + CoT yields relatively moderate results. The marginal improvement in METEOR, modest growth in BLEU-4, and stable performance in ROUGE-1-p all suggest that the step-by-step reasoning mechanism improves the model's expressive quality to some extent. However, compared to the contribution of the KG, the standalone effect of CoT is relatively limited, raising a question worth deeper reflection: to what extent does the effectiveness of a reasoning mechanism depend on adequate knowledge support? The experimental results suggest that

Table 6 Evaluation results—ROUGE

Model	ROUGE								
	1-g			2-g			L-gram		
	f	p	r	f	p	r	f	p	r
Qwen2.5-7B	0.2111	0.1519	0.4076	0.0507	0.0347	0.1251	0.1581	0.1128	0.3160
DeepSeekR1-7B	0.2031	0.1654	0.3305	0.0373	0.0277	0.0822	0.1555	0.1279	0.2540
Mistral-7B	0.2496	0.2653	0.2741	0.0609	0.0582	0.0787	0.2047	0.2145	0.2301
GraphRAG	0.4089	0.4471	0.4316	0.2143	0.2088	0.2469	0.2737	0.3152	0.3299
LightRAG	0.4011	0.4122	0.3911	0.1660	0.1991	0.1607	0.2821	0.3496	0.2765
Vector-RAG	0.4182	0.3722	0.3747	0.1820	0.2110	0.1658	0.2827	0.3002	0.2690
Qwen2.5-7B	0.2111	0.1519	0.4076	0.0507	0.0347	0.1251	0.1581	0.1128	0.3160
Qwen2.5-DFT	0.3068*	0.3513*	0.4571*	0.1445**	0.1821*	0.2242**	0.2672*	0.2518*	0.3141**
Qwen2.5-IFT	0.2663*	0.2006*	0.4589*	0.1195*	0.1121*	0.2058*	0.2158**	0.1590*	0.3085
FT-LLM	0.3921*	**0.4637***	0.3762**	0.2217*	**0.2678****	0.2124*	0.3083*	0.3671**	0.3056*
FT + CoT	0.3868	0.4413	0.3671	0.2232*	0.2310	0.2242*	0.3472*	0.3549	0.3269*
FT + KG	0.4165*	0.4216	0.4503*	0.2056	0.2433	0.2560*	0.3501*	0.3249	0.3523*
PD-LLM	**0.4258***	0.4314	**0.4731***	**0.2457***	0.2509	**0.2878***	**0.3676****	**0.3695***	**0.4114****

Bold value indicates the best-performing result for each metric among all compared models

Table 7 Evaluation results—BLEU, METEOR and GPT4-HFR

Modle	BLEU				METEOR	GPT4-HFR
	1-g	2-g	3-g	4-g		
DeepSeekR1-7B	0.0816	0.377	0.0195	0.0111	0.1935	0.6337
Mistral-7B	0.1618	0.0777	0.0445	0.0283	0.1608	0.5920
Llama3.1-8B	0.0860	0.0413	0.0237	0.0148	0.1707	0.6013
Vector-RAG	0.2364	0.1326	0.1065	0.0760	0.3422	0.7735
GraphRAG	0.2426	0.1296	0.0943	0.0810	0.3449	0.7924
LightRAG	0.1795	0.1320	0.1055	0.0860	0.3519	0.7735
Qwen2.5-7B	0.1023	0.0559	0.0353	0.0247	0.1306	0.6125
Qwen2.5-DFT	0.2162*	0.1563*	0.0896*	0.0688*	0.2963**	0.6981*
Qwen2.5-IFT	0.1927*	0.1373*	0.0778*	0.0697**	0.2626*	0.7169*
FT-LLM	0.2476*	0.1419*	0.0953**	0.0718*	0.3320*	0.6972*
FT + CoT	0.2400	0.1520*	0.0860	0.0743*	0.3328*	0.7022*
FT + KG	0.2431*	0.1465*	0.1015*	0.0821**	0.3465*	0.7846**
PD-LLM	**0.2484****	**0.1520***	**0.1100***	**0.0866***	**0.3521***	**0.8222***

Bold values indicate the best-performing result for each metric among all compared models

optimizing reasoning steps alone has limited impact when knowledge gaps are present, and the depth of reasoning is ultimately constrained by the knowledge the model already possesses.

After introducing explicit KG retrieval and CoT reasoning, the improvements of PD-LLM are mainly reflected in three dimensions: recall, long-range consistency, and language fluency. ROUGE-1-r increased from 0.3762 to 0.4731, and ROUGE-L-r rose from 0.3056 to 0.4114, indicating that external knowledge helps the model capture more semantic entities and their relationships. The increase in ROUGE-L-f (+ 5.9%) suggests that the multi-step reasoning mechanism makes the text structure more coherent. Meteor rose from 0.3320 to 0.3521, showing that semantic preservation and expression diversity have improved simultaneously. Additionally, the 20.6% growth in BLEU-4 further demonstrates the positive effect of chain reasoning in complex sentence combination and high-order dependencies.

From the perspective of hallucination control, the HFR metric provides direct evidence of system reliability. A base model HFR of 0.6125 means that approximately one-third of generated content contains accuracy issues of varying degrees. Two-stage fine-tuning raised this to 0.6972, indicating that injecting domain knowledge does improve the model's factual accuracy, a result of repeated learning from real policy content during the fine-tuning process. The introduction of a KG brought the HFR to 0.7846, with the mechanism behind this improvement being that explicit knowledge structures provide the model with clear factual constraints, rather than merely statistical knowledge probabilities. The complete system PD-LLM achieved an HFR of 0.8222, meaning that more than a quarter of the performance improvement can be attributed to the synergistic effect of the system's components. This significant reduction in hallucinations stems from several key factors in the system design. First, the CoT mechanism makes the reasoning process supervisable and constrainable at intermediate steps, reducing the propagation of errors across steps. Second, the five-stage cascade framework ensures output quality at each stage through phased processing and role definition. Third, the information saturation stopping condition prevents fabricated content that might be introduced by excessive reasoning. The combined effect of these design choices indicates that in professional domain applications, hallucination control is not merely a technical problem but a system design problem, requiring multi-pronged constraints at the architectural level.

Comparison with RAG baseline methods provides insight into architectural design choices. Vector-RAG and GraphRAG represent two different approaches—vector retrieval and graph-structure augmentation—with HFR scores of 0.7735 and 0.7924 respectively. The key to PD-LLM surpassing these methods lies in its dynamic knowledge interaction mechanism—traditional RAG transitions to the generation phase once the retrieval stage is complete, whereas PD-LLM maintains interaction with knowledge sources throughout the entire reasoning process. This difference in design enables the system to dynamically adjust its query strategy when initial knowledge proves insufficient, achieving multi-round progressive problem solving. This characteristic makes PD-LLM better suited to handling highly complex and ambiguous question scenarios in government affairs question answering.

However, while PD-LLM outperformed FT-LLM in the vast majority of metrics, it slightly lagged behind in ROUGE-1-p and ROUGE-2-p (by approximately 3 percentage points). This indicates that although the knowledge injection strategy enhanced information coverage and overall F1, there was a minor loss in local literal accuracy, which can be regarded as a normal trade-off resulting from the expansion of recall.

Although the model exhibits excellent performance in metric-based evaluations, from a linguistic standpoint, none of the three metrics can transcend the surface structure evaluation paradigm put forward by Chomsky [55]. As a result, they are unable to penetrate into the deep structure and pragmatic aspects. Particularly within highly specialized registers such as official government texts, the relationship between language expression and functional realization is far more intricate. It is necessary to take into account dimensions that go beyond mere lexical matching, including legal accuracy, information integrity, and functionality. These limitations not only reveal the inadequacies of existing evaluation methods in achieving a profound understanding of texts within specific domains but also underscore the pressing need for a systematic and multi-dimensional assessment of LLMs within practical application scenarios. This is especially true in specialized contexts, such as the knowledge Q&A function related to democratic demands.

Analysis of generated content

In light of the notable scarcity of systematic research regarding the functional assessment of large language model-based knowledge Q&A systems tailored to democratic demands, and considering that the datasets currently employed for metric-based evaluations are inherently flawed in quality, using these datasets as a reference standard for the content generated by the models may cause the evaluation results to deviate from the models' actual capabilities. Moreover, it could even exacerbate the existing biases in the model outputs. Consequently, this study builds upon the overall framework for evaluating LLMs in the domain of scientific and technological intelligence research, as proposed by Li et al. [56]. It conducts a comprehensive evaluation of the functions of LLM knowledge Q&A systems from four distinct dimensions: fundamental capabilities, dynamic capabilities, specialized capabilities, and comprehensive capabilities. The detailed evaluation findings are presented in supplementary information A. Owing to the limited scale of the original answer set, some of the Q&A results presented in the tables represent the key points extracted from the LLM's responses after appropriate omissions.

Upon a comprehensive analysis of multi-dimensional assessment outcomes, the model showcases distinct performance across various core competencies. In the realm of basic capabilities, the model manifests exceptional semantic comprehension and knowledge extraction abilities. It can accurately apprehend the core content and contextual elements of policy documents like the "Interim Regulations on the Management of Housing Provident Fund in Urban and Rural Areas". Moreover, it can precisely identify and categorize search queries such as "What are the specific implementation details of the housing provident fund policy in urban and rural areas?". Regarding consultation questions such as "What are the talent introduction policies in Nantong City?", the model can systematically search and present key policy provisions from the "Detailed Rules for the Implementation of the Nantong City Talent Introduction Plan". These include the funding support mechanism ranging from 100,000 to 500,000 yuan and the three-year rent subsidy plan, along with other detailed policy information. During the dynamic capability assessment stage, the model demonstrates strong capabilities in extracting policy elements and conducting multi-dimensional analysis. It can perform a three-dimensional analysis of planning-related issues like "Funding Management and Use of Affordable Housing in Nantong Urban Area" from aspects such as the macro environment and the evolution of the policy system. It not only presents the static content of the policies but also clarifies their evolutionary path and logical relationships. Concerning specialized capabilities, the model uses KG technology to accurately identify user needs and conduct targeted retrieval, enabling in-depth information mining and comprehensive analysis within specific domains. Taking the "Policy on the Collection and Management of Housing Provident Fund in Xuzhou City" as an illustrative case, the model is capable of providing precise interpretations within a limited context. This includes not only the policy change of the contribution ratio being adjusted from 5 to 12% but also the institutional logic underlying it. Regarding the comprehensive ability assessment, the model demonstrates a certain level of information integration and implicit knowledge reasoning capabilities. For complex questions such as the "Employment and Entrepreneurship Policies for College Graduates in Chongchuan District, Nantong City," the model can generate content that is not explicitly presented in the original policy texts through the integration of multi-source information. This includes comprehensive information such as the policy of up to 300,000 yuan in startup support.

Compared to the baseline model Qwen2.5-7B, PD-LLM demonstrates clear advantageous characteristics. In terms of information completeness and the depth of professional knowledge, PD-LLM generates responses with higher information density and greater professional depth. These responses not only cover the surface-level content of policies but also provide in-depth analysis of the underlying logic of policy formulation and implementation mechanisms. This enables a more comprehensive understanding of policies,

moving beyond superficial familiarity to a deeper grasp of the principles and operational frameworks that guide policy execution. Regarding structured cognition and representational capabilities, PD-LLM enhances the readability and accessibility of policy texts through the use of a hierarchical framework and logical segmentation. This approach effectively reduces the cognitive load for users of government affairs information. In the area of precise policy text analysis, PD-LLM exhibits superior citation accuracy and terminology interpretation precision. It is capable of accurately extracting and explaining professional terms and policy clauses from regulatory documents, thereby significantly enhancing the authority and reliability of policy dissemination. From an application-oriented perspective, PD-LLM constructs a more sophisticated solution framework. It not only provides detailed policy interpretations but also outlines a complete roadmap from policy formulation to implementation. In understanding local policies, PD-LLM demonstrates strong context-awareness capabilities. It can accurately identify regional differences in policy implementation and clearly articulate the specificities of local policies within the broader national policy framework. This enables a more nuanced and accurate understanding of how local policies adapt or respond to national directives, considering the unique social, economic, and cultural contexts of different regions. When addressing complex problems, PD-LLM showcases exceptional multi-dimensional analytical capabilities. It synthesizes professional knowledge from diverse disciplines such as economics, culture, and law, and integrates this knowledge to formulate comprehensive solutions. These solutions effectively address the complex and interdisciplinary challenges commonly encountered in modern government affairs. These distinctive strengths make PD-LLM highly professional, precise, and practical within knowledge-based question-answering systems in the government affairs domain. Consequently, PD-LLM offers more efficient and effective solutions for intelligent government information processing. Moreover, it lays a solid foundation for professional consultation services in government-citizen interactions, supporting more informed decision-making and improving the overall quality of public governance.

Although PD-LLM demonstrates strong performance across most capability dimensions, several key limitations remain: (1) In scenarios requiring the coordinated analysis of policies across multiple domains, the model shows limitations in cross-domain knowledge integration. As a result, it struggles to fully comprehend the complex interdependencies and mutual influences among different policies. This constraint hinders the model's ability to deliver a comprehensive and nuanced understanding of real-world policy interactions. (2) When addressing questions that require the evaluation and prediction of policy effects, the model's responses largely remain at the surface level of policy interpretation. It lacks in-depth analytical reasoning, which is essential for accurately assessing potential policy outcomes and generating informed projections. This limitation reduces the model's effectiveness in supporting policy-making and strategic decision-making processes. (3) In handling highly individualized and unconventional public demands, the model tends to generate relatively standardized responses. Its capacity for personalization is limited, making it insufficiently responsive to the unique characteristics and specific needs of such diverse requests. This issue may lead to less effective interactions with users and potentially result in dissatisfaction among those seeking tailored solutions. These limitations highlight the need for future research to focus on improving the model's cross-domain reasoning abilities, deepening the level of policy analysis, and enhancing its adaptability in addressing non-standardized public inquiries.

Comparison of manual responses

Supplementary Information B presents examples of question-and-answer sessions in the policy consultation scenario between PD-LLM and human agents. By comparing the response samples from PD-LLM and human government services, it becomes evident that PD-LLM demonstrates a notable advantage in structured presentation. The model systematically divides the housing provident fund loan process into stages such as preparation, acceptance, and approval, which not only improves the efficiency of information retrieval but also enhances the public's understanding of complex administrative procedures. In terms of information comprehensiveness, PD-LLM overcomes traditional limitations related to response length by offering a detailed overview of the entire workflow, including complaint channels, handling mechanisms, and confidentiality measures. This maximizes information density within a limited textual space and reflects the modern governance principle of transparency. Furthermore, its humanized interactive style fosters a user-friendly service image, significantly improving user satisfaction and the overall consultation experience. More importantly, PD-LLM enhances the predictability and visibility of government services by clearly presenting all administrative processing stages, thereby offering an innovative model for government-citizen interaction.

Nonetheless, in practical applications, PD-LLM still exhibits areas that require improvement. At the systemic level, the model faces inherent limitations in understanding and articulating political contexts. Its comprehension and interpretation of ideological discourse systems do not reach the level of precision found in human responses. This suggests that in situations where political nuances and ideological subtleties are critical, PD-LLM may fail to

convey information with the same accuracy and sophistication as human agents. The model's responses are primarily based on fixed information stored within the policy KG, which makes it inflexible when dealing with diverse and evolving scenarios. In contrast to human respondents, who can make real-time judgments and adjustments based on specific contexts, PD-LLM lacks the capability to dynamically adapt its responses to unique situations. According to global labor market research, administrative and secretarial work, as the backbone of government affairs services, faces a high automation exposure rate of approximately 24% [57]. This means that the optimal use case for PD-LLM—rule-based, standardized policy consultation—falls precisely within the range of tasks most susceptible to automation. Conversely, when it comes to complex demands requiring ethical judgment, political sensitivity, and personalized decision-making, the system's limitations are also most pronounced. Moreover, the dynamic adaptability of intelligent systems like PD-LLM remains insufficient when addressing complex and multifaceted scenarios. Additionally, the mechanisms for real-time responses to new policies and emerging situations need further enhancement. In an era marked by rapid policy changes and shifting social dynamics, the system's inability to promptly and effectively respond to new developments may hinder the smooth delivery of government services.

Looking ahead, a promising direction for future development lies in integrating the structured strengths of PD-LLM with the contextual and political sensitivity of human responses. By combining the advantages of both approaches, a more intelligent and responsive government service answering system can be developed. This hybrid system has the potential to significantly enhance the practical value of intelligent government services in complex scenarios and, in turn, improve public satisfaction. Addressing PD-LLM's current limitations in this way also supports the advancement of more effective, user-centric governance in the digital age.

Conclusion

This study constructs a PD-LLM system through real-time bidirectional coupling of LLMs and government KGs via CoT reasoning. Experimental validation demonstrates that PD-LLM achieves significant improvements over baseline models, with multiple evaluation metrics including BLEU-4, ROUGE-L, and METEOR all improving by over 20%, while the hallucination control metric reaches 0.8222, representing an 18.0% improvement over the baseline. The system maintains overall superiority compared to equivalent open-source models and mainstream RAG solutions. The experimental results demonstrate that the coupling mechanism of KGs and CoT reasoning can substantially suppress factual hallucinations in government service scenarios and enhance the quality of generated content, enabling the model to genuinely understand complex policy logic rather than relying on surface-level matching. Through a multi-turn clarification mechanism, the system automatically identifies user intent, precisely locates relevant policy clauses, and effectively reduces policy comprehension deviation and service mismatching risks. This allows limited human resources to be redirected toward complex decision-making and personalized services, thereby improving public information access efficiency and service experience. During the reasoning process, graph topology metrics are employed to dynamically constrain reasoning steps, replacing subjective judgment with quantifiable structural indicators to ensure reasoning complexity is rigorously controllable and hallucination accumulation is suppressed. Furthermore, open-source models through adaptive design can fully achieve production-grade quality in high-risk, high-validation scenarios typical of government services, reducing dependence on closed-source large models. This provides government departments with a controllable technical solution to transition from passive response to proactive precise service, offering significant practical implications for AI empowerment of grassroots governments and small-to-medium-sized departments.

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Data availability Restrictions apply to the datasets: The dataset provided in this article is not readily available as the data is part of ongoing research. Requests to access the dataset should be directed to 1024112415@njupt.edu.cn.

Declarations

Conflict of interest Corresponding authors declare on behalf of all authors that there is no conflict of interest. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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References

1. Zhu Y, Peng Y, Xin L, Zhang X (2025) Public demand and optimization of smart government service platform from the perspective of service life cycle based on the Kano model: a Chinese case study. PLoS ONE 20:e0319707. https://doi.org/10.1371/journal.pone.0319707
2. Zhang Y, Huang H (2024) Influence of citizens' perceived quality of governmental online response on their political trust and continued e-participation intention: a mixed-methods perspective. Chin J Journal Commun. https://doi.org/10.13495/j.cnki.cjjc.2024.07.001
3. Kong X, He Z (2024) How does digital and intelligence technology trust among grassroots civil servants come into being?-Empirical evidence based on the swift response to public complaints reform in Beijing. Chin Publ Adm. https://doi.org/10.19735/j.issn.1006-0863.2024.11.10
4. Bai W (2024) Paradigm innovation of artificial intelligence participation in digital government construction-interactive agile governance based on general large model of artificial intelligence in the administrative domain (GovGPT). J Northwest Minzu Univ (Philos Soc Sci). https://doi.org/10.14084/j.cnki.cn62-1185/c.20240527.010
5. Ren H, Liu J, Wang J et al (2025) Research on intelligent question answering system based on large language model. Comput Eng Appl 61:1. https://doi.org/10.3778/j.issn.1002-8331.2409-0300
6. Shen S, Feng S, Wu N, Zhao Z (2024) Research on retrieval-augmented generation of policy texts based on large language models. Data Anal Knowl Discov 9:37–48
7. Chen Y, Hou X (2024) Reform and development of judicial theory in the era of big data. Jilin Univ J Humanit Soc Sci 64:146–157:234–235. https://doi.org/10.15939/j.jujsse.2024.06.fx3
8. Qian G, Yang Z, Shi L (2025) Reshaping the policy system of AI big model industry: American experience and Chinese path. J Technol Econ 44:14–27
9. Yang J, Kong F (2025) Opening the "Public Black Box": perception and reconstruction of public encounters. J Public Adm 179–195:200
10. Yuan Z, Li L, He F et al (2025) Traditional chinese medicine question answering model based on chain- of- thought and knowledge graph. Comput Eng Appl 61:158–166
11. Edge D, Trinh H, Cheng N et al (2025) From local to global: a graph RAG approach to query-focused summarization. arXiv preprint. arXiv:2404.16130
12. Liu X, Gao H, Shao C et al (2024) Construction and application of large language model for public complaints with knowledge reasoning and similarity retrieval. J Front Comput Sci Technol 18:2940–2953
13. Wei J, Tay Y, Bommasani R et al (2022) Emergent abilities of large language models. Trans Mach Learn Res
14. Zhang Q, Wang Y, Wang H et al (2024) Comprehensive review of large language model fine-tuning. Comput Eng Appl 60:17–33
15. Wang H, Liu C, Xi N, et al (2023) HuaTuo: tuning LLaMA model with Chinese medical knowledge. arXiv preprint. http://arXiv.org/abs/2304.06975
16. Wang N, Yang H, Wang C (2023) FinGPT: instruction tuning benchmark for open-source large language models in financial datasets. arXiv preprint. arXiv:2310.04793
17. Fei Z, Zhang S, Shen X et al (2024) InternLM-law: an open source Chinese legal large language model. arXiv preprint. http://arxiv.org/abs/2406.14887
18. Musumeci E, Brienza M, Suriani V et al (2024) LLM based multi-agent generation of semi-structured documents from semantic templates in the public administration domain. In: Degen H, Ntoa S (eds) Artificial intelligence in HCI. Springer Nature Switzerland, Cham, pp 98–117
19. Zhong Y, Wong D, Lan K (2024) Tax intelligent decision-making language model. IEEE Access 12:146202–146212. https://doi.org/10.1109/ACCESS.2024.3419079
20. Devlin J, Chang M-W, Lee K, Toutanova K (2019) BERT: Pre-training of deep bidirectional transformers for language understanding. In: Burstein J, Doran C, Solorio T (eds) Proceedings of the 2019 conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, volume 1 (long and short papers). Association for Computational Linguistics, Minneapolis, pp 4171–4186
21. Li M, Wang L, Lai H (2024) Identification of China′s S&T policy properties based on deep learning. Kexue Yanjiu Guanli 45:1. https://doi.org/10.19571/j.cnki.1000-2995.2024.02.001
22. Hu Z, Li P, Wang J, Xiong X (2024) Research on policy tools classification based on chatgpt augmentation and supervised contrastive learning. Comput Eng Appl 45:292–305
23. Xiang X, Xiang X, Hu M et al (2024) Research on science and technology policy and regulation Q&A system driven by large models. J Front Comput Sci Technol 18:2349–2360
24. Liu W, Liu Q (2025) Top 10 trends in generative artificial intelligence and response strategies for public cultural institutions. Libr Dev. https://doi.org/10.19764/j.cnki.tsgjs.20250294
25. Pan S, Luo L, Wang Y et al (2024) Unifying large language models and knowledge graphs: a roadmap. IEEE Trans Knowl Data Eng 36:3580–3599. https://doi.org/10.1109/TKDE.2024.3352100
26. Edge D, Trinh H, Cheng N et al (2025) From local to global: A graph RAG approach to query-focused summarization. arXiv preprint. http://arXiv.org/abs/2404.16130
27. Wang J, Chen M, Hu B et al (2024) Learning to plan for retrieval-augmented large language models from knowledge graphs. In: Al-Onaizan Y, Bansal M, Chen Y-N (eds) Findings of the Association for Computational Linguistics: EMNLP 2024. Association for Computational Linguistics, Miami, pp 7813–7835
28. Chen L, Tong P, Jin Z, et al (2024) Plan-on-graph: self-correcting adaptive planning of large language model on knowledge graphs. arXiv preprint. http://arXiv.org/abs/2410.23875
29. Suryawanshi S, Waghmode S, Sawant R, Gupta M (2025) A knowledge graph-based RAG for cross-document information extraction. 2025 5th international conference on pervasive computing and social networking (ICPCSN). IEEE, Salem, pp 1401–1406
30. Zhang T, Liu Y, Li B et al (2022) History-aware hierarchical transformer for multi-session open-domain dialogue system. In: Goldberg Y, Kozareva Z, Zhang Y (eds) Findings of the Association for Computational Linguistics: EMNLP 2022. Association for Computational Linguistics, Abu Dhabi, pp 3395–3407
31. Wei J, Wang X, Schuurmans D et al (2022) Chain of thought prompting elicits reasoning in large language models. arXiv preprint. http://arXiv.org/abs/2201.11903v6

32. Wang L, Xu W, Lan Y et al (2023) Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. In: Rogers A, Boyd-Graber J, Okazaki N (eds) Proceedings of the 61st annual meeting of the Association for Computational Linguistics (volume 1: long papers). Association for Computational Linguistics, Toronto, pp 2609–2634
33. Kong A, Zhao S, Chen H et al (2024) Better zero-shot reasoning with role-play prompting. arXiv preprint. http://arXiv.org/abs/2308.07702
34. Vaidya NN, Runkler TA, Hubauer T et al (2025) Conceptual in-context learning and chain of concepts: solving complex conceptual problems using large language models. In: 2025 IEEE symposium on computational intelligence in natural language processing and social media (CI-NLPSoMe), pp 1–7
35. Liu H, An J, Jia X et al (2025) Stacked intelligent metasurfaces for wireless communications: applications and challenges. IEEE Wirel Commun 32:46–53. https://doi.org/10.1109/MWC.001.2500002
36. Dai Z (2024) Applications and challenges of large language models in smart government—from technological advances to regulated applications. In: Proceedings of the 2024 3rd international conference on frontiers of artificial intelligence and machine learning. ACM, Yichang China, pp 275–280
37. Chen X, Lu Z (2024) Petition: mechanism for the expression of people's demands and the resolution of social contradictions. J Zhejiang Gongshang Univ. https://doi.org/10.14134/j.cnki.cn33-1337/c.2024.05.002
38. Hua B, Kang Y, Fan L (2022) Knowledge modeling and association Q&A for policy texts. Data Anal Knowl Discov 68:79–92
39. Han J, Lu J, Xu Y et al (2024) Intelligent practices of large language models in digital government services. IEEE Access 12:8633–8640. https://doi.org/10.1109/ACCESS.2024.3349969
40. Ji S, Pan S, Cambria E et al (2022) A survey on knowledge graphs: representation, acquisition, and applications. IEEE Trans Neural Netw Learn Syst 33:494–514. https://doi.org/10.1109/TNNLS.2021.3070843
41. Wu M, Yi X, Yu H et al (2022) Nebula graph: an open source distributed graph database. arXiv preprint. arXiv:2206.07278
42. Tang A, Soulier L, Guigue V (2025) Clarifying ambiguities: on the role of ambiguity types in prompting methods for clarification generation. In: Proceedings of the 48th international ACM SIGIR conference on research and development in information retrieval. Association for Computing Machinery, New York, pp 20–30
43. Lewis-Lim S, Tan X, Zhao Z, Aletras N (2026) Can confidence estimates decide when chain-of-thought is necessary for LLMs? arXiv preprint. arXiv:2510.21007
44. Kim H, Lamb TA, Bibi A et al (2025) Detecting LLM hallucination through layer-wise information deficiency: analysis of ambiguous prompts and unanswerable questions. In: Christodoulopoulos C, Chakraborty T, Rose C, Peng V (eds) Proceedings of the 2025 conference on empirical methods in natural language processing. Association for Computational Linguistics, Suzhou, pp 32310–32322
45. Kadavath S, Conerly T, Askell A et al (2022) Language models (mostly) know what they know. arXiv preprint. arXiv:2207.05221
46. Ma X, Zhang G, Li B et al (2025) Policy knowledge graph construction based on multi-factor semantic relevance. Inf Stud Theory Appl
47. Ye C, Duan H, Zhang H et al (2023) Multi-source data repairing: a comprehensive survey. Mathematics (Basel) 11:2314. https://doi.org/10.3390/math11102314
48. Euzenat J, Shvaiko P (2013) Ontology matching. Springer, Berlin, Heidelberg
49. Trivedi H, Balasubramanian N, Khot T, Sabharwal A (2023) Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In: Rogers A, Boyd-Graber J, Okazaki N (eds) Proceedings of the 61st annual meeting of the Association for Computational Linguistics (volume 1: long papers). Association for Computational Linguistics, Toronto, pp 10014–10037
50. Kang M, Kwak JM, Baek J, Hwang SJ (2023) Knowledge graph-augmented language models for knowledge-grounded dialogue generation. https://doi.org/10.48550/ARXIV.2305.18846
51. Hu EJ, Shen Y, Wallis P et al (2021) LoRA: low-rank adaptation of large language models. In: arXiv.org. https://arXiv.org/abs/2106.09685v2. Accessed 10 June 2025
52. Kulkarni A, Zhang Y, Moniz JRA et al (2025) Evaluating evaluation metrics—the mirage of hallucination detection. In: Christodoulopoulos C, Chakraborty T, Rose C, Peng V (eds) Findings of the Association for Computational Linguistics: EMNLP 2025. Association for Computational Linguistics, Suzhou, pp 19013–19032
53. Guo Z, Xia L, Yu Y et al (2025) LightRAG: Simple and fast retrieval-augmented generation. In: Christodoulopoulos C, Chakraborty T, Rose C, Peng V (eds) Findings of the Association for Computational Linguistics: EMNLP 2025. Association for Computational Linguistics, Suzhou, pp 10746–10761
54. Xu M, Fralick D, Zheng JZ et al (2017) The differences and similarities between two-sample T-test and paired T-test. Shanghai Arch Psychiatry 29:184–188. https://doi.org/10.11919/j.issn.1002-0829.217070
55. Chomsky N (2009) Syntactic structures. De Gruyter Mouton
56. Li X, Li Z, Zhao K et al (2024) Reflection on the llms evaluation in the field of S&T information research. Inf Stud Theory Appl 170–176:200. https://doi.org/10.16353/j.cnki.1000-7490.2024.11.018
57. Pokhrel S, Poudel S, Banjade SR et al (2025) Generative AI and job vulnerability: a global review. Recent Res Rev J 4:281–305. https://doi.org/10.36548/rrrj.2025.2.006